Answer Key: Naive Bayes Exercises

Section A: Conceptual Answers

1) Bayes' Theorem:
$P(A|B) = [P(B|A) \times P(A)] / P(B)$

2) The term 'naive' refers to the assumption of conditional independence among features.

3) Conditional Independence:
$P(X1, X2, ..., Xn \mid C) = \Pi\, P(Xi \mid C)$

Section B: Mathematical Solutions

1) Spam Example:
$P(D) = 0.70(0.30) + 0.20(0.70) = 0.35$
$P(Spam \mid Discount) = 0.60$

2) Weather Example:
$P(Yes) = 4/6 = 0.667$
$P(Sunny \mid Yes) = 1/4 = 0.25$
$P(Sunny \mid No) = 2/2 = 1$
Prediction: No

3) Gaussian Naive Bayes (Score = 65):
Posterior $P(A|65) \approx 0.613$
Posterior $P(B|65) \approx 0.387$
Prediction: Class A